Md. Abdur Rahman Shibly

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Dhaka, Bangladesh

Education

Bangladesh University of Engineering and Technology (BUET) *March 2018 – May 2023*
Bachelor of Science
CGPA: 3.29 / 4.00
BSc Thesis:
Building a Semantic Search Engine using an Ontology: A Case Study on Social Media Influencers.

Research Interests

- Applied AI
- LLM
- NLP
- Conversational AI
- Virtual Reality
- RAG
- Software Engineering
- Hallucination Reduction
- HCI
- Ontology and Knowledge Graph

Ongoing Research

Semantic Search Engine using an Ontology *January 2023 - Current*
Tools: Protege, Neo4j, Django, Python, NumPy, Pandas [Supervisor]
Developed a novel ontology for social media influencers to categorize them by audience location, interaction rates, and content types. Current work focuses on refining assumptions from prior research by fine-tuning LLaMA2 and utilizing prompt engineering techniques.

Online Facility Assignment with Nonuniform Capacities *April 2025 - Current*
Tools: Python, Pandas, NumPy, SciPy, Matplotlib [Supervisor]
Investigating the online facility assignment problem with nonuniform facility capacities in metric spaces. Established a reduction from the general setting to the unit-capacity case, simplifying the analysis of greedy algorithms. Derived tighter competitive ratio bounds for algorithms on Euclidean planes and unweighted graphs, extending prior results on uniform capacities.

Publications

Conference Papers

1. Shibly, M. A. R., et al., "Leveraging the Domain Adaptation of Retrieval-Augmented Generation Models for Question Answering," in *Proceedings of the 22nd International Conference on Information Technology – New Generations (ITNG 2025)*, Springer, 2025. DOI

Book Chapters

1. Chapter in Springer Nature book collection based on ITNG 2025 conference proceedings. Link

Test Scores

IELTS (August 2025): Overall Band **8.0** (Reading: 9, Listening: 8, Speaking: 8, Writing: 6.5)

Professional Experience

Software Engineer *March 2025 – Present*
Bengal Mobile QA Solutions — Dhaka, Bangladesh

- Developed AI-powered systems, including AI Ruler, a dynamic parser for structuring raw CDRs
- Built chatbots and personalized AI assistants using LLMs for hotel interactions and parent-focused applications
- Led development of Invoice-Factory-AI, applying prompt engineering to extract structured data from invoices
- Modernized hotel management system with microservices-based architecture for scalability and maintainability

Assistant Software Engineer *July 2023 – February 2025*
Computer Network Systems — Dhaka, Bangladesh

- Co-led government insurance project BISDP-BI, overseeing data analysis and ML model development for intelligent decision-making and automation
- Collaborated on AI-driven projects including Aptitudo and Zeuxis.AI, contributing to model development and data analysis
- Worked with cross-functional teams to design, test, and integrate AI/ML components into production-level solutions

Consultant *December 2023 – July 2024*
Bengal Mobile QA Solutions — Dhaka, Bangladesh

- Developed AI assistant for hotel industry using LLMs, focusing on conversational AI solutions tailored for hotel-specific tasks
- Applied expertise in LLM-based systems for creating specialized industry applications

Projects

BISDP-BI *August 2023 – August 2024*
Government Insurance Project
Developed comprehensive government insurance project focusing on data analysis and ML model development for intelligent decision-making regarding fraudulent claims and monetary predictions. Co-led the machine learning team throughout the entire project lifecycle.
Python, Pandas, NumPy, Scikit-learn, Machine Learning Models

Invoice-Factory-AI *May 2025 – July 2025*
AI-Powered Invoice Processing
Developed AI system using LLM and prompt engineering to parse hotel career invoices of various formats, extracting structured data automatically. Integrated Google Document AI for OCR capabilities.
Django, LLM, Prompt Engineering, Google Document AI

Parental Assistant *August 2025 – Present*
Multi-Agent AI System
Building comprehensive AI assistant for parents to help monitor and support their children using advanced multi-agent system architecture.
Langgraph, Langchain, Langfuse, Multi-agent Systems

Zeuxis.AI *July 2024 – December 2024*
Multimedia Generation Platform
Developed comprehensive platform enabling users to generate images and videos from text or image inputs, including face swapping capabilities in a unified interface.
Java, Spring Boot, FAL AI, Vision Models

Aptitudo *May 2024 – August 2024*
AI Interview Assistant
Created AI-powered interview preparation system generating diverse question types including MCQ, coding, and written questions, with automated scoring capabilities. [Demo Video]
Django, LLM, Prompt Engineering

References

Dr. Muhammad Masroor Ali
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Bangladesh University of Engineering and Technology
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Relationship: Thesis Advisor

Dr. Md. Mostofa Akbar
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Relationship: Research Supervisor